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# Zero-Divisor-Preserving $q$ -Richardson Extrapolation for the Fabius–Rvachev Sinc Product

Exact same-parity nesting thresholds, holomorphic continuation through  
zeros,  
and certified local spectral jets

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A frontier report based on a recursive audit of all 68 \*.tex sources under Analysis/FabiusFunction/docs, including current expositions, active frontier drafts, archived revisions, Fourier-decay audits, and transcribed source papers. The report starts where the latest repository drafts stop: it resolves their open same-parity nesting problem and removes the logarithmic extrapolator's apparent obstruction at zeros by factoring the finite zero divisor before extrapolating the zero-free tail.

Research synthesis prepared for Vladimir Reshetnikov

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## Abstract

Let

$$P_b(t) = \prod_{k \geq 1} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad \operatorname{sinc} z = \frac{\sin z}{z}, \quad b > 1,$$

and put  $q = b^{-2}$ . The inspected repository already develops exact logarithmic  $q$ -Richardson extrapolation for finite sections of this product. Its transform has alternating error sign and a complete Bernoulli–zeta/Bell-polynomial expansion, but the logarithmic definition is stated only away from the zeros; the same source leaves monotonicity between extrapolants of the same parity as an open parameter-classification problem.

Both gaps are closed here. For the prefix  $P_{b,N}$ , the omitted tail is zero-free in the natural disk  $|t| < \pi b^{N+1}$ . We extrapolate analytic logarithms of the *relative partial tails* and then multiply by  $P_{b,N}$ . The resulting factorized transform is holomorphic across every prefix zero, preserves the complete zero divisor with multiplicity, and agrees with the earlier multiplicative transform wherever the latter is defined. Its quotient by  $P_b$ , after cancellation of common zeros, is exactly  $\exp(E_{N,r})$ , where

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m \geq r+1} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} \left( \frac{t}{\pi b^{N+1}} \right)^{2m}.$$

Consequently, the alternating brackets survive at zeros as certified brackets for the first nonzero derivative, and Cauchy estimates certify every coefficient of the zero-normalized local germ.

Write  $M_r(q, a) = |E_{N,r}|$  after replacing the final even power by  $a^m$ ,  $0 < a < 1$ . The exact classification proved here is

$$M_{r+2}(q, a) < M_r(q, a) \quad \text{for every } 0 < a < 1 \iff q^{r+1}(1+q) \leq 1.$$

If the inequality fails, the two same-parity errors cross exactly once. The proof is coefficientwise: the shared-coefficient ratio is strictly increasing in  $m$  and tends to  $q^{2r+3}/((1-q^{r+1})(1-q^{r+2}))$ . Thus all levels are nested from order zero exactly when  $q \leq (\sqrt{5}-1)/2$ , or equivalently  $b \geq \sqrt{\varphi}$ . In the dyadic Fabius–Rvachev case  $q = 1/4$ , every level is nested. This gives two monotone analytic chains squeezing the Fourier product while preserving all included integer zeros, their multiplicities  $1 + \nu_2(n)$ , their Thue–Morse lobe signs, and their first nonzero derivatives.

Numerical experiments, supplied as a fully commented Python program, independently compare direct products with the exact coefficient series, compute the threshold curve and unique crossings, and demonstrate derivative-ratio errors below  $10^{-24}$  at the multiplicity-five zero  $x = 48$  by order eight. The final sections formulate conjectures and a formalization program for general refinement masks, complex boundary continuation, optimal finite precision, inverse-Fabius coupling, and periodically perturbed Lambert saddles.

## Proof and novelty status

**Established inputs from the repository.** The Fabius/Rvachev identities; exact dyadic arithmetic; Thue–Morse,  $q$ -binomial, Bernoulli, Bell, Lambert- $W$ , and inverse asymptotic machinery; the infinite sinc product and its integer zero multiplicities; the  $q$ -Lagrange weights; the exact transformed logarithmic tail series; alternating error sign; bounded weight condition number; phase-locked endpoint Richardson transforms; and the large Fourier-decay theory are treated in the inspected corpus.

**Proved in this report.** The factorized zero-divisor-preserving transform; its holomorphic extension through all included zeros; exact zero-normalized quotient and local jet identities; complex nonasymptotic bounds; the necessary-and-sufficient same-parity nesting criterion; uniqueness and analyticity of the supercritical crossing; the  $\sqrt{\varphi}$  base threshold; large-order threshold asymptotics; and the nested dyadic spectral-jet brackets.

**Not claimed.** “New” means not found in the recursively inspected repository or in the targeted primary-source searches described in [Section 1.4](#). It is not an unqualified assertion of worldwide priority. None of the new theorems is claimed to have a Lean counterpart at the snapshot commit.

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# 1 Corpus boundary, audit method, and collision control

## 1.1 Pinned recursive corpus

The analysis is pinned to commit 7f4e6e6fdd57f756f303b20f56b5382627e5e1d4. This matters because the active frontier subtree was changing during the work: the GitHub code-search index lagged the tree, while the recursive Git objects gave the exact count of 68 TeX files. Their complete path manifest is reproduced in Section A and supplied as a machine-readable text file in the archive.

| Stratum                          | Files | Audit role                                                                                                                                                                    |
|----------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Living expositions and synthesis | 5     | Formalization-backed baseline, glossary, non-elementarity, Lean map, and consolidated frontier volume.                                                                        |
| Active frontier drafts           | 24    | Arithmetic rays, half-integer spectra, inverse asymptotics, Pascal–Rvachev hierarchies, zero determinants, $q$ -Richardson analysis, and twelve Fourier-decay reports/audits. |
| Frozen archive                   | 32    | Revision history, early notes, superseded expositions, prior $q$ -calculus articles, convergence studies, and source provenance.                                              |
| Transcribed source papers        | 7     | Arias de Reyna, Aistleitner–Hofer–Larcher, Toth, Allouche–Shallit, and related primary context.                                                                               |
| Total                            | 68    | Every recursive <code>*.tex</code> path at the pinned commit.                                                                                                                 |

Archived files were not treated as 32 independent current papers. They were read as a revision graph: distinctive claims were checked against the current descendant and the archive’s own consolidation/provenance maps. The twelve Fourier documents were similarly read as two waves plus two comparative audits, so a superseded constant or conjecture could not be mistaken for the current state of the problem.

## 1.2 The mature baseline

The corpus is unusually dense. Among other things, it already contains:

1. a proof-oriented primary exposition of the bounded Fabius function, Rvachev’s compact up-function, and the signed global extension, with exact separation of those objects;
2. probability, convolution, spline, Fourier, moment, exact dyadic,  $q$ -binomial, and Thue–Morse representations;
3. a quadratic-plus-periodic negative-Laplace exponent, lower-branch Lambert phase, all-orders endpoint and inverse expansions, closed inverse jets, and Bell-polynomial coefficient transfer;
4. finite geometric-grid Richardson filters with exact Gaussian  $q$ -binomial weights, alternating logarithmic brackets, complete transformed tail coefficients, and bounded condition number;
5. integer-zero spectral determinants, exact multiplicities, half-integer spectral arithmetic, Prouhet quadrature, rational dyadic rays, cyclotomic orbit norms, and a Pascal–Rvachev hierarchy;
6. a two-wave analysis of pointwise, metric, extremal, quadratic-mean, Cesàro, and shell-adaptive Fourier decay.

The main challenge was therefore not producing another list of identities. It was finding a statement that survives comparison with the newest same-day drafts.

## 1.3 Candidates deliberately discarded

Several promising-looking ideas were rejected after exact comparison:

- phase-locked reciprocal-grid Richardson extrapolation for the Lambert saddle was already present, including its complete homogeneous and Bell-polynomial moments;
- the integer zero divisor, spectral zeta, heat trace, local all-order zero jets, and  $q$ -deformation already had a dedicated report;
- arithmetic rational rays and half-integer weighted Prouhet formulae had newer active treatments;
- generic log-periodic Fourier envelopes and transfer-operator constants were covered by the comparative audits;
- the  $q$ -Pochhammer/Thue–Morse dictionary for endpoint corrections and Fourier-zero multiplicities was already developed in the spectral-endpoint report.

### The two surviving openings

The latest  $q$ -Richardson draft explicitly leaves two questions open: whether same-parity brackets are monotone in identifiable parameter regimes, and how the logarithmic transform should be continued across zeros. The present report resolves both questions exactly.

## 1.4 External literature boundary

The external search was used to prevent reinvention of standard extrapolation theory and to locate adjacent product-approximation work. Generalized Richardson theory on geometric meshes, including convergence and stability, is classical [9, 10]. The  $q$ -binomial identities used to write the weights belong to standard basic hypergeometric calculus [8]. The Lambert function and its branch asymptotics are standard [11]. The Fabius/Rvachev baseline comes from Arias de Reyna and related work [2, 3, 4], while the lacunary product’s metric theory is represented by Aistleitner–Hofer–Larcher [5].

Targeted searches located recent Padé work for products of functions, but not a construction that factors a finite canonical zero divisor and applies geometric  $q$ -Richardson extrapolation to the zero-free infinite tail, nor the exact threshold in theorem 4.3. The nearest classical results concern extrapolation of asymptotic scales, not preservation of an entire function’s moving zero divisor.

## 2 Established logarithmic $q$ -Richardson structure

### 2.1 The lacunary sinc product

Fix  $b > 1$  and set

$$q = b^{-2} \in (0, 1). \quad (2.1)$$

We use the entire normalized sinc function

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases}$$

and define

$$P_b(t) = \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{t}{b^k}\right), \quad P_{b,N}(t) = \prod_{k=1}^N \operatorname{sinc}\left(\frac{t}{b^k}\right). \quad (2.2)$$

Uniform convergence on compact sets makes  $P_b$  entire. The first omitted factor has its nearest zeros at  $\pm \pi b^{N+1}$ , so the natural tail disk is

$$D_N = \{t \in \mathbb{C} : |t| < \pi b^{N+1}\}. \quad (2.3)$$

The complete and partial relative tails are

$$T_N(t) = \frac{P_b(t)}{P_{b,N}(t)} = P_b\left(\frac{t}{b^N}\right), \quad T_{N,j}(t) = \frac{P_{b,N+j}(t)}{P_{b,N}(t)}, \quad j \geq 0, \quad (2.4)$$

with  $T_{N,0} = 1$ .

**Lemma 2.1** (Zero-free tail disk). *Every  $T_{N,j}$ , including  $T_N$ , is holomorphic and zero-free on  $D_N$ . It therefore has a unique holomorphic logarithm on  $D_N$  normalized to vanish at  $t = 0$ . On  $D_N \cap \mathbb{R}$ , all these tails are positive.*

*Proof.* Every zero of  $\text{sinc}(t/b^k)$  has modulus at least  $\pi b^k$ . For  $k \geq N + 1$  this is at least the radius of  $D_N$ , with equality only for the first omitted factor and only on the boundary. Thus every finite relative tail and the locally uniform infinite tail are zero-free in the disk. The disk is simply connected, giving the normalized logarithm. For real  $|t| < \pi b^{N+1}$  every omitted argument has modulus less than  $\pi$ , where  $\sin x/x > 0$ .  $\square$

## 2.2 Bernoulli–zeta coefficients

For  $|z| < \pi$ ,

$$-\log \text{sinc } z = \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} z^{2m} = \sum_{m=1}^{\infty} \frac{(-1)^{m+1} 2^{2m-1} B_{2m}}{m(2m)!} z^{2m}. \quad (2.5)$$

Consequently, with

$$u_N(t) = \frac{t}{\pi b^{N+1}}, \quad (2.6)$$

one has the exact prefix error

$$\text{Log } P_{b,N}(t) - \text{Log } P_b(t) = \sum_{m \geq 1} \frac{\zeta(2m)}{m(1 - q^m)} u_N(t)^{2m} \quad (2.7)$$

whenever both logarithms are taken on a zero-free component. The right side itself is holomorphic throughout  $D_N$  and will be the invariant object used below.

## 2.3 Geometric Lagrange weights

For  $r \geq 0$ , let  $\lambda_{r,j}(q)$  be the Lagrange interpolation weights at nodes  $1, q, \dots, q^r$ , evaluated at the origin:

$$\lambda_{r,j}(q) = \frac{(-1)^{r-j} q^{(r-j)(r-j+1)/2}}{(q; q)_j (q; q)_{r-j}}, \quad 0 \leq j \leq r, \quad (2.8)$$

where  $(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k)$ . Their transformed moments are

$$S_{r,m}(q) = \sum_{j=0}^r \lambda_{r,j}(q) q^{jm} = \begin{cases} 1, & m = 0, \\ 0, & 1 \leq m \leq r, \\ (-1)^r \frac{q^{r(r+1)/2} (q^{m-r}; q)_r}{(q; q)_r}, & m \geq r + 1. \end{cases} \quad (2.9)$$

This is the finite  $q$ -binomial cancellation mechanism already developed in the repository.

Away from zeros, the existing multiplicative transform can be written schematically as

$$\exp \left( \sum_{j=0}^r \lambda_{r,j} \text{Log } P_{b,N+j} \right).$$

Its exact logarithmic error and alternating sign are established there. The obstruction is that no single logarithm of  $P_{b,N+j}$  can pass through a zero. The next section removes that obstruction without changing the transform where it was already defined.

### 3 Factorization through the complete zero divisor

#### 3.1 Definition and holomorphic continuation

**Definition 3.1** (Factorized tail transform). On  $D_N$ , let  $\text{Log } T_{N,j}$  be the normalized logarithms from theorem 2.1. Define

$$\widehat{T}_{N,r}(t) = \exp \left( \sum_{j=0}^r \lambda_{r,j}(q) \text{Log } T_{N,j}(t) \right), \quad (3.1)$$

$$\widehat{P}_{b,N,r}^{\text{fac}}(t) = P_{b,N}(t) \widehat{T}_{N,r}(t). \quad (3.2)$$

The superscript “fac” will usually be suppressed once no confusion with the unfactored logarithmic expression is possible.

**Theorem 3.2** (Zero-divisor-preserving continuation). *For every  $N, r \geq 0$ :*

1.  $\widehat{P}_{b,N,r}^{\text{fac}}$  is holomorphic on  $D_N$ .
2. Wherever  $P_{b,N}$  is nonzero, it agrees with the earlier multiplicative  $q$ -Richardson transform formed from  $P_{b,N}, \dots, P_{b,N+r}$ .
3.  $P_b, P_{b,N}$ , and  $\widehat{P}_{b,N,r}^{\text{fac}}$  have exactly the same zero divisor in  $D_N$ , including multiplicities.
4. After cancellation of that common zero divisor, the quotient  $\widehat{P}_{b,N,r}^{\text{fac}}/P_b$  extends to a holomorphic zero-free function on all of  $D_N$ .

*Proof.* The function  $\widehat{T}_{N,r}$  is an exponential of a holomorphic function and is therefore holomorphic and zero-free. This proves the first assertion after multiplication by the finite prefix.

If  $P_{b,N}(t) \neq 0$ , then

$$\text{Log } P_{b,N+j} = \text{Log } P_{b,N} + \text{Log } T_{N,j}$$

on the locally normalized branch. Since  $\sum_j \lambda_{r,j} = 1$ , the unfactored weighted logarithm equals  $\text{Log } P_{b,N} + \sum_j \lambda_{r,j} \text{Log } T_{N,j}$ , proving agreement.

By theorem 2.1, both the exact tail  $T_N$  and the extrapolated tail  $\widehat{T}_{N,r}$  are zero-free in  $D_N$ . Thus every zero of either full product comes from  $P_{b,N}$  with unchanged multiplicity. Finally,

$$\frac{\widehat{P}_{b,N,r}^{\text{fac}}}{P_b} = \frac{\widehat{T}_{N,r}}{T_N}$$

after cancellation, and the right side is holomorphic and zero-free.  $\square$

*Remark 3.3.* The construction is not a numerical patch at a real zero. It is a holomorphic factorization on the full complex disk  $D_N$ . The finite prefix is a canonical carrier of the entire zero divisor visible at scale  $N$ ; only the zero-free determinant tail is extrapolated.

#### 3.2 Exact error on the whole disk

**Theorem 3.4** (Exact zero-normalized logarithmic error). *On  $D_N$ ,*

$$\frac{\widehat{P}_{b,N,r}^{\text{fac}}(t)}{P_b(t)} = \exp(E_{N,r}(t)) \quad (3.3)$$

after cancellation at common zeros, where

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} u_N(t)^{2m}. \quad (3.4)$$



In particular,

$$E_{N,r}(t) = (-1)^r \frac{q^{r(r+1)/2} \zeta(2r+2)}{(r+1)(1-q^{r+1})} u_N(t)^{2r+2} + O(u_N(t)^{2r+4}). \quad (3.5)$$

*Proof.* From (2.7), the logarithm of the  $j$ th relative partial tail is

$$\text{Log } T_{N,j} = \text{Log } T_N + \sum_{m \geq 1} \frac{\zeta(2m)}{m(1-q^m)} u_N(t)^{2m} q^{jm}.$$

Weighting and using  $\sum_j \lambda_{r,j} = 1$  gives

$$\text{Log } \frac{\widehat{T}_{N,r}}{T_N} = \sum_{m \geq 1} \frac{\zeta(2m)}{m(1-q^m)} u_N(t)^{2m} S_{r,m}(q).$$

Insert (2.9). The first  $r$  powers vanish, and the remaining closed form is (3.4). At  $m = r+1$  one has  $(q; q)_r = (q^{m-r}; q)_r$ , giving (3.5).  $\square$

### 3.3 Uniform complex bounds

**Proposition 3.5** (A computable disk bound). *Let  $0 < R < \pi b^{N+1}$  and put*

$$a_R = \left( \frac{R}{\pi b^{N+1}} \right)^2.$$

*Then on  $|t| \leq R$ ,*

$$|E_{N,r}(t)| \leq B_{N,r}(R) := \frac{q^{r(r+1)/2}}{(q; q)_r} \frac{\zeta(2)}{(1-q)(r+1)} \frac{a_R^{r+1}}{1-a_R}. \quad (3.6)$$

*Consequently,*

$$\left| \frac{\widehat{P}_{b,N,r}^{\text{fac}}}{P_b} - 1 \right| \leq e^{B_{N,r}(R)} - 1 \quad (3.7)$$

*after zero cancellation, uniformly on the closed disk.*

*Proof.* In (3.4), use  $(q^{m-r}; q)_r \leq 1$ ,  $(1-q^m)^{-1} \leq (1-q)^{-1}$ ,  $\zeta(2m) \leq \zeta(2)$ , and

$$\sum_{m \geq r+1} \frac{a_R^m}{m} \leq \frac{a_R^{r+1}}{(r+1)(1-a_R)}.$$

Then use  $|e^z - 1| \leq e^{|z|} - 1$ .  $\square$

### 3.4 Local jets at every included zero

**Theorem 3.6** (Exact spectral-jet quotient). *Let  $t_0 \in D_N$  be a zero of  $P_b$  of multiplicity  $\mu$ . Then*

$$\frac{(\widehat{P}_{b,N,r}^{\text{fac}})^{(\mu)}(t_0)}{P_b^{(\mu)}(t_0)} = \exp(E_{N,r}(t_0)). \quad (3.8)$$

*More strongly, after removing the common factor  $(t - t_0)^\mu$ , the quotient of the two zero-normalized germs is exactly  $\exp(E_{N,r}(t))$ . Hence for every  $k \geq 1$ ,*

$$\left( \frac{d}{dt} \right)^k \text{Log} \left( \frac{\widehat{P}_{b,N,r}^{\text{fac}}(t)/(t - t_0)^\mu}{P_b(t)/(t - t_0)^\mu} \right) = E_{N,r}^{(k)}(t) \quad (3.9)$$

*in a neighborhood of  $t_0$ .*

*Proof.* Write  $P_{b,N}(t) = (t - t_0)^\mu A(t)$  with  $A(t_0) \neq 0$ . Then

$$P_b(t) = (t - t_0)^\mu A(t)T_N(t), \quad \widehat{P}_{b,N,r}^{\text{fac}}(t) = (t - t_0)^\mu A(t)\widehat{T}_{N,r}(t).$$

Taking the first nonzero derivative gives the tail ratio; [theorem 3.4](#) identifies that ratio with  $\exp(E_{N,r}(t_0))$ . Cancelling the complete common germ gives the stronger identity, and logarithmic differentiation gives (3.9).  $\square$

**Corollary 3.7** (Cauchy-certified higher coefficients). *Choose  $\delta > 0$  so that the closed disk  $|t - t_0| \leq \delta$  lies in  $D_N$ . Every Taylor coefficient of the zero-normalized quotient is bounded explicitly by applying Cauchy's estimate to  $\exp(E_{N,r}) - 1$  and [theorem 3.5](#) on that disk. In particular, the entire local jet converges uniformly as either  $N \rightarrow \infty$  at fixed  $r$  or  $r \rightarrow \infty$  at fixed interior scale.*

### 3.5 Bernoulli and Bell transfer at zeros

The exponent (3.4) is a Bernoulli–zeta series. Write locally

$$E_{N,r}(t) = \sum_{m \geq r+1} e_{r,m} u_N(t)^{2m}.$$

Then

$$\exp(E_{N,r}(t)) = 1 + \sum_{n \geq r+1} \frac{1}{n!} B_n(1!x_1, 2!x_2, \dots, n!x_n) u_N(t)^{2n}, \quad (3.10)$$

where  $x_m = e_{r,m}$  for  $m \geq r+1$  and  $x_m = 0$  below that order, and  $B_n$  is the complete exponential Bell polynomial. Bell transfer itself is established in the repository; the new point is that (3.10) now applies directly to the ratio of the first nonzero derivatives and to every zero-normalized local germ. No separate limiting argument at a zero is required.

## 4 Exact classification of same-parity nesting

### 4.1 Positive error magnitudes

For  $0 < a < 1$ , define

$$M_r(q, a) = \frac{q^{r(r+1)/2}}{(q; q)_r} \sum_{m=r+1}^{\infty} \frac{\zeta(2m)}{m} \frac{(q^{m-r}; q)_r}{1 - q^m} a^m. \quad (4.1)$$

Thus on the real tail disk,  $E_{N,r} = (-1)^r M_r(q, u_N(t)^2)$ . The established alternating bracket only compares opposite sides of the target. Comparing  $r$  with  $r+2$  is subtler.

**Lemma 4.1** (Shared-coefficient ratio). *For  $m \geq r+3$ , the ratio of the  $a^m$  coefficient in  $M_{r+2}$  to that in  $M_r$  is*

$$R_{r,m}(q) = \frac{q^{2r+3}(1 - q^{m-r-2})(1 - q^{m-r-1})}{(1 - q^{r+1})(1 - q^{r+2})}. \quad (4.2)$$

For fixed  $q, r$ , this is strictly increasing in  $m$  and

$$R_{r,\infty}(q) = \frac{q^{2r+3}}{(1 - q^{r+1})(1 - q^{r+2})}. \quad (4.3)$$

Moreover,

$$R_{r,\infty}(q) \leq 1 \iff q^{r+1}(1 + q) \leq 1. \quad (4.4)$$

*Proof.* The prefactor ratio is

$$\frac{q^{(r+2)(r+3)/2}/(q; q)_{r+2}}{q^{r(r+1)/2}/(q; q)_r} = \frac{q^{2r+3}}{(1 - q^{r+1})(1 - q^{r+2})}.$$

The quotient of shifted  $q$ -Pochhammer factors contributes the two factors in the numerator of (4.2); all remaining factors, including  $\zeta(2m)/m$  and  $1 - q^m$ , cancel. Monotonicity and the limit are immediate. Finally,

$$q^{2r+3} \leq (1 - q^{r+1})(1 - q^{r+2}) = 1 - q^{r+1} - q^{r+2} + q^{2r+3}$$

is equivalent to  $q^{r+1}(1 + q) \leq 1$ .  $\square$

## 4.2 Boundary asymptotics

**Lemma 4.2** (Accelerated boundary decomposition). *Let*

$$A_r(q) = \frac{q^{r(r+1)/2}}{(q; q)_r}, \quad U_{r,m}(q) = \frac{(q^{m-r}; q)_r}{1 - q^m}.$$

Then

$$\frac{M_r(q, a)}{A_r(q)} = -\log(1 - a) - \sum_{m=1}^r \frac{a^m}{m} + H_r(q, a), \quad (4.5)$$

where

$$H_r(q, a) = \sum_{m \geq r+1} \frac{\zeta(2m)U_{r,m}(q) - 1}{m} a^m \quad (4.6)$$

is analytic in a disk strictly larger than the unit disk. In particular,

$$M_r(q, a) = A_r(q) \log \frac{1}{1 - a} + O_{q,r}(1) \quad (a \uparrow 1). \quad (4.7)$$

*Proof.* The factor  $U_{r,m}(q)$  approaches one with error  $O_{q,r}(q^m)$ , while  $\zeta(2m) - 1 = O(4^{-m})$ . Thus the coefficient in (4.6) decays geometrically. Splitting one from  $\zeta(2m)U_{r,m}$  leaves the harmonic tail  $\sum_{m \geq r+1} a^m/m$ , which is the first two terms in (4.5).  $\square$

The decomposition is also the stable numerical formula used in the supplied code; it avoids summing a nearly harmonic series when  $a$  is close to one.

## 4.3 Necessary and sufficient condition

**Theorem 4.3** (Complete same-parity nesting classification). *Fix  $q \in (0, 1)$  and  $r \geq 0$ . Then*

$$M_{r+2}(q, a) < M_r(q, a) \quad \text{for every } a \in (0, 1) \quad (4.8)$$

if and only if

$$\boxed{q^{r+1}(1 + q) \leq 1}. \quad (4.9)$$

If (4.9) fails, there is a unique number  $a_r(q) \in (0, 1)$  such that

$$\begin{cases} M_{r+2}(q, a) < M_r(q, a), & 0 < a < a_r(q), \\ M_{r+2}(q, a) = M_r(q, a), & a = a_r(q), \\ M_{r+2}(q, a) > M_r(q, a), & a_r(q) < a < 1. \end{cases} \quad (4.10)$$

The crossing is simple, and  $a_r(q)$  is real-analytic in  $q$  throughout the supercritical region  $q^{r+1}(1 + q) > 1$ .

*Proof.* Suppose first that (4.9) holds. By theorem 4.1,  $R_{r,m}(q) < 1$  for every finite shared index  $m \geq r + 3$ ; the inequality is strict even at the limiting threshold. Therefore every shared coefficient of  $M_{r+2}$  is strictly smaller than its counterpart in  $M_r$ , while  $M_r$  also has the two additional positive coefficients at  $m = r + 1, r + 2$ . This proves (4.8) coefficientwise.

Conversely, if the threshold fails, then  $A_{r+2}(q)/A_r(q) = R_{r,\infty}(q) > 1$ . By theorem 4.2,

$$M_r(q, a) - M_{r+2}(q, a) = (A_r(q) - A_{r+2}(q)) \log \frac{1}{1-a} + O(1),$$

which tends to  $-\infty$  as  $a \uparrow 1$ . Near  $a = 0$ , the difference is positive because its leading term is the  $a^{r+1}$  term present only in  $M_r$ . Thus at least one crossing exists.

For uniqueness, theorem 4.1 shows that the coefficient difference has exactly one sign transition: positive coefficients first, followed eventually by negative coefficients. Let  $K$  be the final index with nonnegative coefficient. If  $D(a) = M_r(q, a) - M_{r+2}(q, a) = \sum c_m a^m$ , then

$$\frac{d}{da}(a^{-K}D(a)) = \sum_m (m-K)c_m a^{m-K-1} < 0,$$

because  $(m-K)c_m \leq 0$  on both sides of the transition, with at least one strict term. Hence  $a^{-K}D(a)$  is strictly decreasing and has at most one zero. The crossing is simple; at its zero, the displayed derivative equals  $a^{-K}D'(a) < 0$ . The implicit-function theorem then gives real analyticity in  $q$ .  $\square$

*Remark 4.4.* The theorem is stronger than a sufficient small-tail estimate. It identifies the exact parameter boundary, and when that boundary is crossed it describes the only possible failure mode: one and only one reversal near the first omitted zero.

#### 4.4 Golden-ratio threshold and eventual nesting

**Corollary 4.5** (All levels nested). *Same-parity nesting holds for every order  $r \geq 0$  if and only if*

$$q(1+q) \leq 1, \tag{4.11}$$

that is,

$$q \leq \frac{\sqrt{5}-1}{2} = \varphi^{-1}, \quad \text{or equivalently} \quad b \geq \sqrt{\varphi} = 1.2720196495\dots \tag{4.12}$$

*Proof.* The left side of (4.9) decreases with  $r$ , so order zero is the only binding constraint. Solve  $q^2 + q - 1 \leq 0$  and use  $q = b^{-2}$ .  $\square$

**Corollary 4.6** (First uniformly nested order). *For every  $q \in (0, 1)$ , same-parity nesting is automatic from the finite order*

$$r_*(q) = \max \left\{ 0, \left\lceil \frac{\log(1+q)}{-\log q} \right\rceil - 1 \right\} \tag{4.13}$$

onward.

Thus even a geometric base arbitrarily close to one eventually enters the nested regime.

#### 4.5 Threshold curve and its asymptotics

Let  $q_r$  be the unique solution of

$$q_r^{r+1}(1+q_r) = 1. \tag{4.14}$$

The first values are shown in Table 1.

**Proposition 4.7** (Large-order threshold). *Put  $n = r + 1$  and  $L = \log 2$ . Then*

$$q_r = 1 - \frac{L}{n} + \frac{L(L+1)}{2n^2} - \frac{L(4L^2 + 15L + 6)}{24n^3} + O(n^{-4}), \tag{4.15}$$

$$b_r = q_r^{-1/2} = 1 + \frac{L}{2n} + \frac{L(L-2)}{8n^2} + O(n^{-3}). \tag{4.16}$$

*Proof.* Insert  $q = 1 + c_1 n^{-1} + c_2 n^{-2} + c_3 n^{-3} + \dots$  into  $n \log q + \log(1+q) = 0$  and compare powers of  $n^{-1}$ . This gives  $c_1 = -L$ ,  $c_2 = L(L+1)/2$ , and  $c_3 = -L(4L^2 + 15L + 6)/24$ . The binomial expansion of  $q^{-1/2}$  gives the second formula.  $\square$

Table 1: Exact same-parity threshold  $q_r$  and equivalent base  $b_r = q_r^{-1/2}$ .

| $r$ | $q_r$             | $b_r$            |
|-----|-------------------|------------------|
| 0   | 0.618033988749895 | 1.27201964951407 |
| 1   | 0.754877666246693 | 1.15096392525776 |
| 2   | 0.819172513396164 | 1.10487288165008 |
| 3   | 0.856674883854503 | 1.08041842739812 |
| 4   | 0.881271461633570 | 1.06523431150218 |
| 5   | 0.898653712628699 | 1.05488183427278 |

## 5 The dyadic Fabius–Rvachev product

### 5.1 Normalization and integer zeros

Under the Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ , write

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}(\pi x / 2^h), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (5.1)$$

This is the  $b = 2$  case, so  $q = 1/4$ . Its nonzero zeros are precisely the integers, with

$$\text{ord}_{x=m} \Phi = 1 + \nu_2(|m|), \quad m \in \mathbb{Z} \setminus \{0\}. \quad (5.2)$$

The sign on  $(m, m+1)$  is the signed Thue–Morse value  $(-1)^{s_2(m)}$  for  $m \geq 0$ ; these are established inputs from the corpus.

Let

$$\Phi_N(x) = \prod_{h=0}^N \text{sinc}(\pi x / 2^h), \quad \mathcal{T}_N(x) = \frac{\Phi(x)}{\Phi_N(x)} = \Phi\left(\frac{x}{2^{N+1}}\right). \quad (5.3)$$

The tail is zero-free and positive on  $|x| < 2^{N+1}$ . Define the factorized transforms  $\widehat{\Phi}_{N,r}$  exactly as in theorem 3.1, using relative partial tails and the weights at  $q = 1/4$ .

### 5.2 Two monotone zero-preserving chains

**Theorem 5.1** (Dyadic nested brackets). *For every  $N \geq 0$  and every real  $|x| < 2^{N+1}$ ,*

$$\widehat{\mathcal{T}}_{N,1}(x) < \widehat{\mathcal{T}}_{N,3}(x) < \cdots < \mathcal{T}_N(x) < \cdots < \widehat{\mathcal{T}}_{N,4}(x) < \widehat{\mathcal{T}}_{N,2}(x) < \widehat{\mathcal{T}}_{N,0}(x), \quad (5.4)$$

*with the evident finite interpretation. Multiplication by  $|\Phi_N(x)|$  gives the same nested inequalities for absolute values of the full products. Every transform has exactly the same integer zeros and multiplicities as  $\Phi$  inside the interval.*

*Proof.* The exact error is  $(-1)^r M_r(1/4, a)$  with  $a = (x/2^{N+1})^2$ . Opposite parities lie on opposite sides of one. Since  $(1/4)^{r+1}(1 + 1/4) < 1$  for every  $r$ , theorem 4.3 makes each parity monotone. Zero preservation follows from theorem 3.2.  $\square$

**Corollary 5.2** (Certified derivatives at all included integer zeros). *Let  $m \neq 0$  be an integer with  $|m| < 2^{N+1}$ , and put  $\mu = 1 + \nu_2(|m|)$ . Then*

$$\frac{\widehat{\Phi}_{N,r}^{(\mu)}(m)}{\Phi^{(\mu)}(m)} = \exp\left((-1)^r M_r\left(\frac{1}{4}, \frac{m^2}{2^{2N+2}}\right)\right). \quad (5.5)$$

*Hence the odd orders give increasing lower bounds and the even orders decreasing upper bounds for  $|\Phi^{(\mu)}(m)|$ . The sign of the derivative is reproduced exactly.*

This supplies a certified local spectral evaluator at every integer zero. Earlier zero-jet formulae in the repository are exact symbolic identities; the present result is complementary: it gives monotone finite-product approximants with an explicit nonasymptotic error bound.

### 5.3 Exact preservation of Thue–Morse lobe signs

Choose  $N$  so that a compact real interval lies inside  $(-2^{N+1}, 2^{N+1})$ . All factorized tails are positive there. Therefore the sign of every  $\widehat{\Phi}_{N,r}$  is carried entirely by the finite prefix  $\Phi_N$ , exactly as for  $\Phi$ . In particular, on each lobe  $(m, m+1)$ ,

$$\operatorname{sgn} \widehat{\Phi}_{N,r}(x) = \operatorname{sgn} \Phi(x) = (-1)^{s_2(m)}. \quad (5.6)$$

Thus the extrapolation does not merely place zeros correctly; it preserves the automatic sign coding between them.

### 5.4 Low-order dyadic weights

The first filters are

$$\begin{aligned} r = 0 : & \quad (1), \\ r = 1 : & \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \\ r = 2 : & \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \\ r = 3 : & \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \end{aligned}$$

Their bounded total variation, proved in the source draft, explains why high order remains numerically usable in the dyadic case. The new nesting theorem adds a separate structural fact: increasing the order by two can never worsen the exact real-axis bracket anywhere inside the first omitted zero.

### 5.5 Adaptive global evaluation

**Algorithm 5.3** (Zero-safe evaluator). To evaluate  $\Phi$  and its local zero-normalized germ on a compact set  $K \subset \mathbb{C}$ :

1. Choose  $N$  so that  $K \subset \{|x| < 2^{N+1}\}$ .
2. Evaluate the finite prefix  $\Phi_N$  directly, retaining its zeros symbolically or with exact integer factor detection.
3. Form the positive/analytic relative tails  $\mathcal{T}_{N,j}$  for  $0 \leq j \leq r$ .
4. Apply the  $q = 1/4$  weights to their logarithms and exponentiate.
5. Multiply by the prefix. Use (3.6) for a complex disk certificate, or the nested real chains for a two-sided certificate.

The method is adaptive in  $N$  and  $r$ . Increasing  $N$  enlarges the exactly represented zero divisor and reduces the tail variable; increasing  $r$  cancels additional Bernoulli–zeta powers. Near a zero, no subtraction or logarithm of a small full product occurs.

## 6 Structural connections and extensions

### 6.1 $q$ -binomial interpolation is the common finite engine

The exact threshold in [theorem 4.3](#) is independent of the zeta factor. It comes entirely from ratios of the Gaussian  $q$ -binomial Lagrange filters. The special functions do two different jobs:

- Bernoulli numbers and zeta values encode the analytic logarithm of one sinc factor;
- $q$ -Pochhammer symbols encode cancellation across geometric scales.

Their combination makes the error coefficients positive up to the global sign  $(-1)^r$ , which is what turns an extrapolation formula into a rigorous bracket.

**Proposition 6.1** (Universal sufficient criterion for positive logarithmic masks). *Suppose an even analytic factor  $f$  satisfies*

$$-\operatorname{Log} f(z) = \sum_{m \geq 1} c_m z^{2m}, \quad c_m \geq 0,$$

*in a disk about zero, and form the same geometric infinite product and  $q$ -Richardson transform. Then  $q^{r+1}(1+q) \leq 1$  is sufficient for same-parity coefficientwise nesting at order  $r$ , independently of the values of the nonnegative coefficients  $c_m$ .*

*Proof.* The common factor  $c_m$  cancels from the coefficient ratio, leaving exactly (4.2). The proof of the sufficient half of theorem 4.3 applies verbatim.  $\square$

Necessity requires information about the boundary growth of the coefficient series. For sinc, the  $1/m$  tail supplied by  $\zeta(2m) \rightarrow 1$  makes the criterion exact.

## 6.2 Canonical products and spectral determinants

The corpus identifies  $\Phi$  with a dyadically self-similar Fredholm determinant and gives its weighted integer canonical product. In that language,  $\Phi_N$  is a finite collection of spectral layers, while  $\mathcal{T}_N$  is a zero-free determinant tail on the resolvent disk preceding its first eigenvalue. Definition 3.1 can therefore be read as a determinant-preserving extrapolation: the finite spectrum is exact, and only the regular determinant of the unresolved spectral tail is accelerated.

This viewpoint suggests applications to the Pascal–Rvachev hierarchy, whose products have higher dyadic multiplicity weights. The zero factor should again be separated first; the remaining logarithm has positive coefficients with polynomial scale multiplicities, leading to multivariate or confluent  $q$ -Richardson filters.

## 6.3 Connection to inverse-Fabius and Lambert phases

The inverse-Fabius endpoint expansion in the corpus is organized around a lower-branch Lambert coordinate with a smooth periodic perturbation. Its phase-locked reciprocal-grid Richardson transform is already present, so it is not repeated here. The useful new principle is instead modular:

$$\text{factor exact singular/zero structure} \quad + \quad \text{extrapolate only the regular tail.} \quad (6.1)$$

For the Fourier product, the exact structure is a canonical zero divisor. For inverse endpoint saddles, it is the Lambert branch and the periodic phase. A hybrid algorithm can therefore preserve both: exact Lambert phase locking in the outer inversion, and exact integer-zero factoring in every Fourier amplitude entering the coefficient calculation.

## 6.4 What the theorem does not say

The nesting theorem concerns exact arithmetic. It does not assert that arbitrarily high  $r$  is optimal in floating-point arithmetic. The dyadic weights have bounded total variation, but the relative-tail logarithms become increasingly correlated and small; roundoff, argument reduction, and evaluation cost eventually dominate. Choosing the precision-aware optimal order remains a separate numerical-analysis problem.

## 7 Numerical experiments

### 7.1 Reproducibility

The archive contains code `frontier_experiments.py`. It uses 60-decimal-digit `mpmath` arithmetic and performs the following independent checks:

1. direct verification of all finite  $q$ -Lagrange moment identities used in the tables;
2. direct zero-free tail products versus the exact Bernoulli–zeta transformed series;
3. bisection of the exact threshold equation and the unique supercritical crossing;
4. derivative-ratio brackets at a multiple integer zero;
5. generation of every CSV table and PNG figure in this report.

The slowly convergent harmonic part of  $M_r$  is removed analytically by [theorem 4.2](#); only a geometrically convergent correction is summed.

### 7.2 Dyadic nesting across the full tail disk

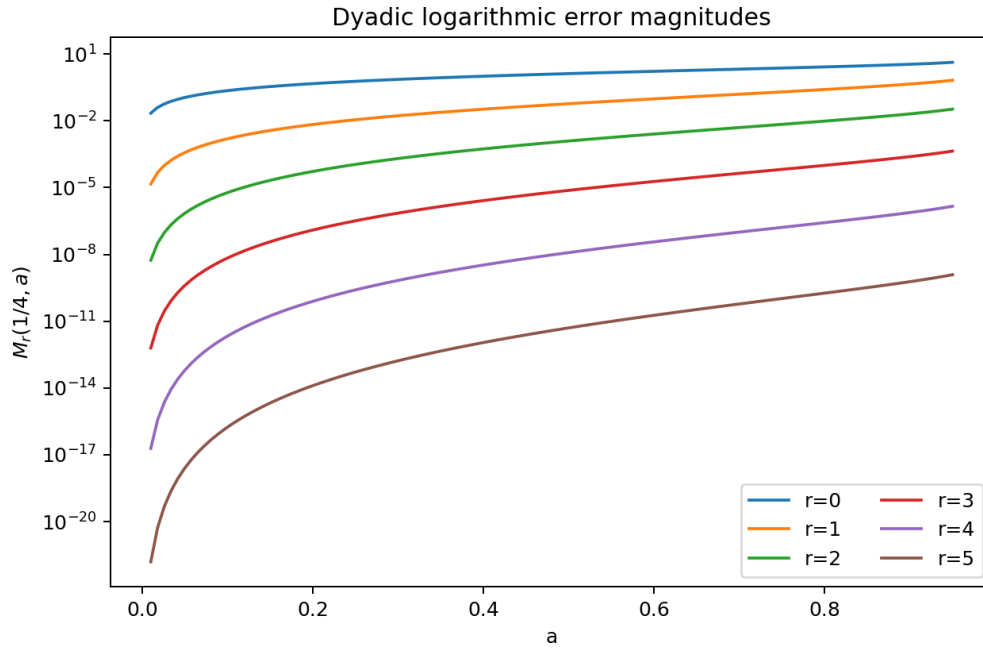


Figure 1: The dyadic magnitudes  $M_r(1/4, a)$  for  $0.01 \leq a \leq 0.95$ . Each parity is strictly nested over the whole interval, as guaranteed by [theorem 4.3](#). The vertical scale is logarithmic.

The separation is dramatic: cancellation order increases by two powers of  $a$  each time, while the prefactor contributes the additional supergeometric factor  $q^{r(r+1)/2}$ .

### 7.3 Unique reversal beyond the threshold

Representative crossing values are



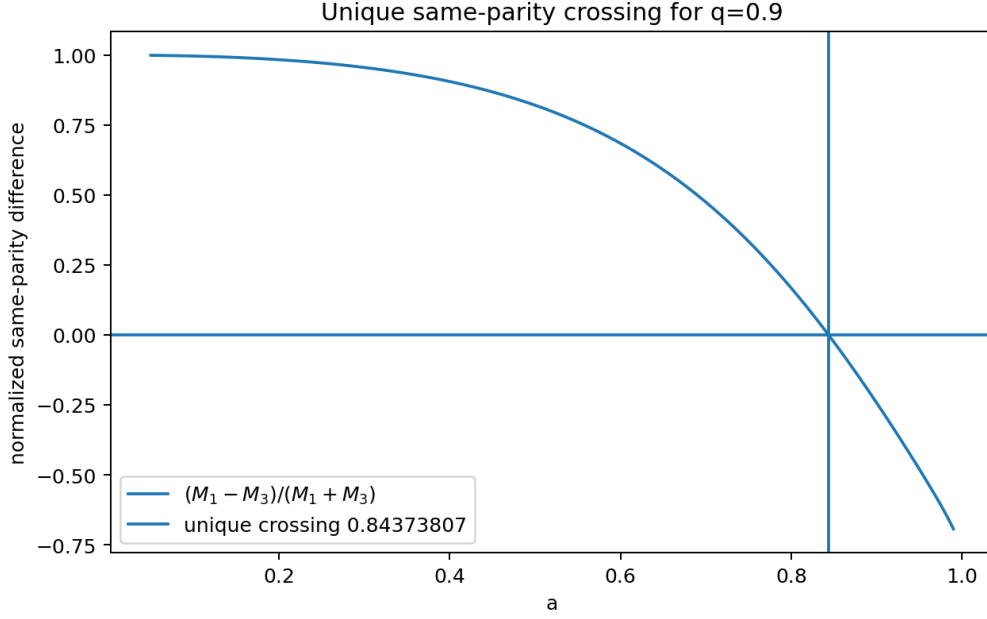


Figure 2: For  $q = 0.9$  and  $r = 1$ , uniform nesting fails. The normalized difference between  $M_1$  and  $M_3$  crosses zero exactly once, at  $a_1(0.9) = 0.843738074718284\dots$

| $q$ | $r$ | $a_r(q)$             |
|-----|-----|----------------------|
| 0.7 | 0   | 0.999706476977084941 |
| 0.8 | 0   | 0.982411424014939233 |
| 0.8 | 1   | 0.980062142113792208 |
| 0.9 | 0   | 0.938630662551898552 |
| 0.9 | 1   | 0.843738074718284012 |
| 0.9 | 2   | 0.847999350512777102 |
| 0.9 | 3   | 0.887485076384088108 |

The roots are numerical illustrations only; existence and uniqueness are proved without numerics.

#### 7.4 A multiplicity-five zero

Take  $m = 48$ , whose zero multiplicity is  $1 + \nu_2(48) = 5$ , and choose  $N = 5$ , so  $|m| < 2^6$ . The exact positive tail is

$$\mathcal{T}_5(48) = \Phi(48/64) = 0.217723205308944050125343\dots$$

By (5.5), every tail ratio below is exactly the ratio of the first nonzero derivative of the approximation to  $\Phi^{(5)}(48)$ .

| $r$ | derivative ratio                | signed log error            | side  |
|-----|---------------------------------|-----------------------------|-------|
| 0   | 4.592987681680617293            | +1.524530723363272884       | upper |
| 1   | 0.922841469541933824            | -8.0297814873516 $10^{-2}$  | lower |
| 2   | 1.001952802143219852            | +1.9508979037801 $10^{-3}$  | upper |
| 3   | 0.999986570110615447            | -1.3429979566325 $10^{-5}$  | lower |
| 4   | 1.000000024414975206            | +2.4414974907466 $10^{-8}$  | upper |
| 5   | 0.999999999988534678            | -1.1465321627915 $10^{-11}$ | lower |
| 6   | 1.000000000000001377            | +1.3766098031980 $10^{-15}$ | upper |
| 8   | 1.00000000000000000000000000325 | +3.24694 $10^{-25}$         | upper |

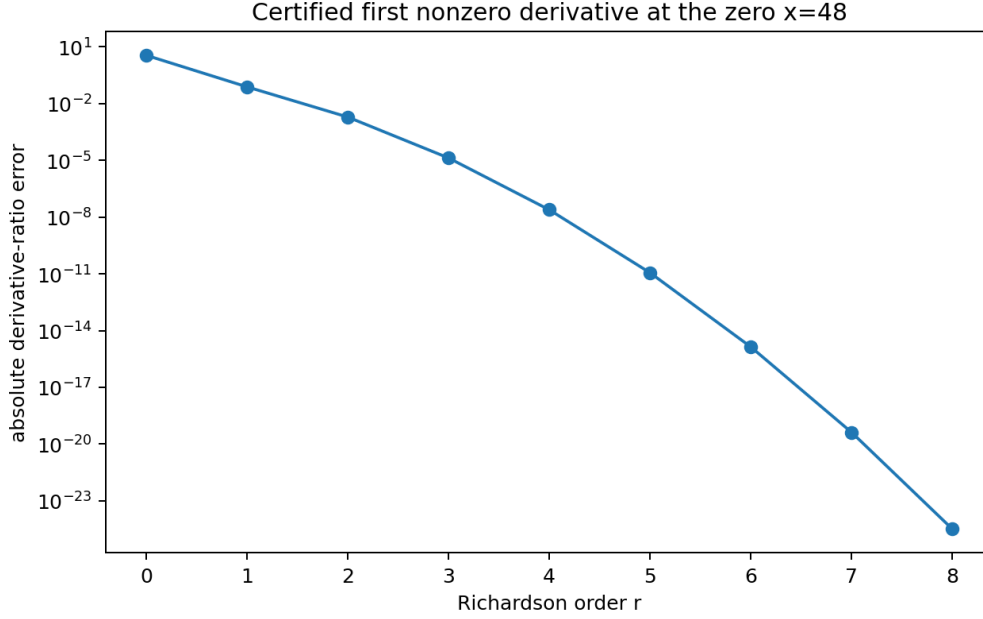


Figure 3: Absolute error in the certified first-nonzero-derivative ratio at the integer zero  $x = 48$ . The factorized construction is well-conditioned here because no logarithm of the vanishing full product is taken.

## 8 Conjectures and frontier directions

The two exact theorems above close specific open items in the current repository, but they also expose several more structured questions.

**Conjecture 8.1** (Regular-variation universality). *Let an even analytic refinement factor satisfy  $-\text{Log } f(z) = \sum_{m \geq 1} c_m z^{2m}$  with  $c_m > 0$  and  $c_m \sim C \rho^{-m}/m$  for some  $C, \rho > 0$ . After normalizing the tail variable so that the first logarithmic singularity occurs at  $a = 1$ , the exact same-parity classification remains  $q^{r+1}(1+q) \leq 1$ , with one unique crossing in the supercritical regime.*

The sinc proof strongly suggests this: coefficientwise sufficiency is already universal by theorem 6.1, and regular variation should reproduce the logarithmic boundary term needed for necessity.

**Conjecture 8.2** (Monotonicity of the crossing curve). *For each fixed  $r$ , the supercritical crossing  $a_r(q)$  decreases from 1 as  $q \downarrow q_r$  and has a finite, strictly positive limit as  $q \uparrow 1$ . A useful sharpening would determine the two endpoint asymptotic expansions explicitly.*

The first limit follows qualitatively from compact convergence and the threshold theorem; the monotonicity and the  $q \uparrow 1$  scale remain to be proved. The numerical values show that dependence on  $r$  is not simply monotone, so a two-parameter saddle analysis is likely needed.

**Problem 8.3** (Optimal finite-precision order). *Given  $q$ , tail radius  $a$ , and working precision  $p$ , determine the order  $r$  minimizing a rigorous sum of truncation, logarithm-evaluation, argument-reduction, and rounding errors. For  $q = 1/4$ , the exact weight condition number remains bounded, so one expects an optimal order growing roughly like a square root of  $p$  after the supergeometric  $q^{r(r+1)/2}$  factor is balanced against finite-precision noise.*

**Problem 8.4** (Complex boundary continuation). *The factorized transform is optimal inside the first omitted zero circle. Extend it across that boundary by moving the newly encountered canonical factors from the tail into the prefix. The goal is an analytic continuation algorithm with overlapping disks, explicit*

transition maps, and uniform Cauchy certificates on arbitrary compact subsets of  $\mathbb{C}$ .

This should be feasible because adjacent factorizations differ by finitely many exact sinc factors. The delicate point is choosing normalized logarithms coherently on overlaps.

**Problem 8.5** (Pascal–Rvachev hierarchy). *For the hierarchy whose Fourier products carry Pascal weights at dyadic scales, construct confluent or multivariate  $q$ -Richardson filters that cancel the polynomial-in-scale multiplicity structure. Determine whether their same-parity thresholds factor into simple inequalities analogous to  $q^{r+1}(1+q) \leq 1$ .*

**Problem 8.6** (Inverse-Fabius coupling). *Use zero-preserving Fourier brackets as certified inputs to the inverse-Fabius saddle calculation. The target is a fully rigorous end-to-end interval algorithm for inverse values in which Lambert phase, periodic corrections, Fourier amplitudes, and all local zero jets carry compatible error enclosures.*

**Conjecture 8.7** (Resurgent organization of phase coefficients). *The all-orders inverse endpoint coefficients, after exact phase locking, admit a Borel transform whose nearest singularities are controlled by the same dyadic complex lattice that governs the Mellin poles and zero-divisor heat trace. If true, optimal truncation and exponentially improved remainders would be determined by one shared spectral geometry.*

**Problem 8.8** (Automatic arithmetic of extrapolated jets). *At integer and half-integer arguments, investigate whether suitably normalized coefficients of the factorized approximants form 2-regular or 2-automatic arrays. The finite prefix has explicit Thue–Morse/Pascal arithmetic, while the tail weights are Gaussian  $q$ -binomials at  $q = 1/4$ ; their interaction may produce new exact recurrences.*

## 9 Lean formalization blueprint

The proof dependencies are unusually clean. A sensible formalization order is:

1. finite  $q$ -Pochhammer algebra and the closed weights (2.8);
2. the transformed-moment identity (2.9);
3. positivity and monotonicity of  $R_{r,m}$ ;
4. the algebraic equivalence (4.4);
5. coefficientwise sufficiency;
6. the accelerated boundary decomposition (4.5);
7. the one-sign-change power-series lemma and unique crossing;
8. analytic zero-free logarithms on a disk;
9. the factorized transform and common-zero-divisor theorem;
10. specialization to the existing Rvachev Fourier product and integer multiplicity API.

The same-parity theorem is mostly finite algebra plus positive series and should be the easier half. The factorized continuation needs complex analysis but avoids branch ambiguity by using a simply connected disk and logarithms normalized at zero. Existing Lean declarations for the Rvachev product, scaling, and integer zeros should make the final specialization short.

## 10 Conclusion

The mature Fabius-function corpus leaves little room for novelty by relabeling familiar identities. The productive route was to take two explicit open defects in its newest  $q$ -Richardson analysis seriously.

Factoring the finite prefix changes the analytic status of the extrapolator: it becomes a holomorphic, zero-divisor-preserving approximation on the entire natural disk, not merely a formula between zeros. The error series survives unchanged, so all Bernoulli, Bell, and  $q$ -binomial structure transfers directly to local spectral jets. Separately, comparing same-parity coefficient arrays yields a complete and unexpectedly simple threshold,  $q^{r+1}(1+q) \leq 1$ , together with a one-crossing theorem when it fails.

For the dyadic Rvachev product the consequence is especially clean. Since  $q = 1/4$  lies far inside the golden-ratio threshold, every order participates in two monotone analytic chains. They squeeze the product, preserve every included integer zero and its Thue–Morse lobe sign, and certify the first nonzero derivative at multiple zeros to arbitrarily high precision. This turns the existing extrapolation machinery into a global spectral tool and provides a compact target for formal verification.

## A Complete recursive TeX manifest

The following manifest is included verbatim from `corpus_manifest.txt`. The first four metadata lines are followed by the 68 unique paths used in the audit.

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**Repository:** VladimirReshetnikov/ProveIt

**Pinned commit:** 7f4e6e6fdd57f756f303b20f56b5382627e5e1d4

**Subtree:** Analysis/FabiusFunction/docs

**TeX files:** 68

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Analysis/FabiusFunction/docs/FabiusFunction\_Mathematical\_Glossary/FabiusFunction\_Mathematical\_Glossary.tex

Analysis/FabiusFunction/docs/Fabius\_Function\_and\_Rvachev\_Up/Fabius\_Function\_and\_Rvachev\_Up.tex

Analysis/FabiusFunction/docs/Non\_Elementarity\_of\_the\_Fabius\_Function/Non\_Elementarity\_of\_the\_Fabius\_Function.tex

Analysis/FabiusFunction/docs/fabius\_lean\_walkthrough/fabius\_lean\_walkthrough.tex

Analysis/FabiusFunction/docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex

Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Fabius\_Arithmetic\_Rays\_Frontier\_Report/Fabius\_Arithmetic\_Rays\_Frontier\_Report.tex

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Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Fabius\_Inverse\_Frontier\_Report\_Source\_and\_PDF/Fabius\_Inverse\_Frontier\_Report.tex

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**B Reproduction files**

The ZIP archive accompanying this report contains:

| Path                                              | Purpose                                                    |
|---------------------------------------------------|------------------------------------------------------------|
| <code>fabius_frontier_report.tex</code>           | Complete LaTeX source.                                     |
| <code>fabius_frontier_report.pdf</code>           | Rendered report.                                           |
| <code>code/frontier_experiments.py</code>         | Fully commented numerical experiment and figure generator. |
| <code>data/same_parity_thresholds.csv</code>      | Threshold roots $q_r$ and bases $b_r$ .                    |
| <code>data/same_parity_crossings.csv</code>       | Representative unique crossing roots.                      |
| <code>data/dyadic_q_richardson_weights.csv</code> | Low-order geometric Lagrange weights.                      |
| <code>data/integer_zero_jet_brackets.csv</code>   | Certified derivative-ratio experiment at $x = 48$ .        |



| Path                                     | Purpose                                                        |
|------------------------------------------|----------------------------------------------------------------|
| data/nonzero_point_factorized_errors.csv | Direct product/series cross-check away from zeros.             |
| figures/*.png                            | The three figures embedded above.                              |
| corpus_manifest.txt                      | Pinned recursive source inventory.                             |
| corpus_manifest_table.tex                | Breakable LaTeX rendering of the inventory used by Appendix A. |
| experiment_summary.txt                   | Compact numerical run summary.                                 |
| requirements.txt                         | Python dependency bounds.                                      |
| README.md                                | Reproduction and file guide.                                   |
| SHA256SUMS                               | SHA-256 checksums for packaged files.                          |

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